

Varun Sahni

LLM Systems, Multi-Agent GenAI, On-Device AI

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EDUCATION

M.S., Computer Science, *New York University, Courant Institute of Mathematical Sciences* Aug. 2026 – May 2028 (Expected)
Focus: Machine Learning, Large Language Models, AI Systems.

B.E., Computer Science, *BITS-Pilani, Pilani Campus* 2020 – 2024
Coursework: Data Structures & Algorithms, Operating Systems, Compilers, Computer Networks, Databases, Computer Architecture. GPA: **8.11/10**

EXPERIENCE

Software Engineer, GenAI / LLM Systems July 2024 – Present
Qualcomm Hyderabad, India

- Architected **Code Buddy**, a multi-agent GenAI code-review platform deployed across **2,500+** repos for **1,400+** engineers, with reviewer, test-author, and security agents under a tool-calling orchestrator.
- Cut RAG inference latency by **75%** by building a Tree-sitter pipeline with hybrid embedding plus symbolic retrieval and Redis-stream caching.
- Scaled the platform to **60,000+** monthly automated reviews, saving an estimated **100,000+** engineer-hours.
- Built prompt-evaluation harnesses on golden datasets to score agent precision/recall, enabling safe iteration across model upgrades.
- Built an agentic tool that reasons over C/C++ modem source via a 10-tool calling loop to drive interactive 4G/5G NAS protocol tests autonomously.
- Shipped **SlideGen**, an LLM tool with structured-output validation and template-aware layout for corporate-compliant decks.

Software Development Intern, Performance / Systems Jan. 2024 – Jun. 2024
Advanced Micro Devices (AMD) Bangalore, India

- Cut bus-trace analysis runtime from a **week to 45 minutes** by building a multi-threaded LSF pipeline that streamed and sharded **250 GB+** datasets.
- Automated triage of **1,000+** JIRA issues via the REST API with hierarchical linkages and auto-summaries.
- Drove hardware-selection decisions by benchmarking CPUs and servers with Sysbench and SPEC CPU 2017 on throughput and tail latency.
- Authored reusable benchmark suites and automation scripts that standardized the team's performance-analysis workflow across platforms.

Software Development Intern May 2023 – Jul. 2023
Standard Chartered GBS Pvt Ltd Chennai, India

- Hardened the **SC Pay** backend with **100+** JUnit and Mockito tests, achieving **98.61%** line coverage.
- Migrated **10+** repositories from Jenkins to Azure DevOps with CI/CD; earned the **CCIB SCPay 2023 Award**.

SELECTED PROJECTS

CabinAI, Hybrid Multi-Agent VLA for Automotive In-Cabin Intelligence 🌐 2026
MediaPipe, Ollama (Qwen-7B INT4), Qwen3-VL-32B, gpt-oss-20b, Qualcomm AIC100, BGE-small ONNX, ChromaDB SparQ 2026 Hackathon

- Architected a **7-agent hybrid edge-cloud vision-language-action system** for real-time driver monitoring: 30 FPS MediaPipe face-mesh DMS and Qwen-7B INT4 (**40 tok/s**) handle safety and voice on-device, escalating complex reasoning to Qwen3-VL-32B / gpt-oss-20b on Qualcomm AIC100.
- Routed **80%+** of queries to edge via a sub-1 ms rule-based router, with multi-tier fallback (AIC100, QGenie Claude, mock) keeping every agent online.
- Built a **15-minute-ahead** proactive fatigue forecast at **500 ms** by sending a **13.5 KB** INT8 buffer plus the latest cabin frame to a Qwen3-VL-32B VLM.
- Powered local RAG over a 31-document corpus in **<50 ms** via BGE-small ONNX (no PyTorch); preempts non-safety bus events when drowsiness **> 0.7**.

MyAI, Privacy-First Personal On-Device AI Assistant 🌐 2026

Kotlin, Android (SDK 35), Qualcomm Genie SDK + QNN v2.45, Hexagon NPU v79, Qwen2.5-1.5B / Llama-3.2-3B INT4 SparQ 2026 Hackathon

- Designed a **4-tier on-device inference router** (trie, cache, tools, INT4 LLM) that resolves most queries without loading the full LLM.
- Ran Qwen2.5-1.5B / Llama-3.2-3B INT4 at **~30 tok/s** (w4a16) on Hexagon NPU v79 via the Qualcomm Genie SDK and QNN runtime.
- Enforced **kernel-level privacy** by omitting the INTERNET permission and gating it with a Gradle CI task (assertNoInternet), verified zero egress on-device.
- Built an on-device agent / tool layer (calculator, datetime, calendar, contacts, alarm) that the LLM invokes via structured JSON tool-calls.
- Added a feedback-driven learning cache that, after one thumbs-up, serves a repeat query in **<10 ms** instead of **~600 ms** of model inference.

HealthSense, On-Wearable Multimodal Biosignal Foundation Model 🌐 2026

PyTorch, ONNX Runtime + QNN EP, Hexagon NPU, INT8 quantization, PhysioNet MIT-BIH AFib SparQ 2026 Hackathon

- Trained **PanLUNA**, a **5.4M-param** multimodal Transformer foundation model that fuses five wearable biosignals (EEG, ECG, PPG, EMG, IMU) into a shared 1024-d latent, then routes it to 4 task heads detecting cardiac arrhythmia, sleep stage, stress, and EEG abnormality from one forward pass.
- Ran INT8 inference on Hexagon NPU at **~95 ms / 19 mJ per pass** with a one-line ONNX hot-swap between real and synthetic cardiac models.
- Validated the cardiac head on **PhysioNet MIT-BIH AFib** (530 windows, 4 records): **96.8%** AFib sensitivity, **91.7%** sinus specificity.
- Built **BioTrain**, an on-device fine-tuning loop that personalises the model to each wearer (last-2-block backprop + GroupNorm in 0.67 MB RAM), lifting stress-detection F1 from **0.78 to 0.91** in **~3.6 s** under **50 mW** while keeping all raw biosignals on the device.

Compiler Construction (C) 🌐 Feb. 2023 – Apr. 2023
Academic Project BITS Pilani

- Built an end-to-end compiler (DFA twin-buffer lexer, LL(1) parser with FIRST/FOLLOW gen, AST, codegen) cutting parse-tree memory by **70%**.
- Added semantic analysis with type checking and scope resolution, plus panic-mode error recovery for graceful handling of malformed input.

ACHIEVEMENTS

- SparQ Qualcomm TechFest 2025 Finalist**: presented the Code Buddy multi-agent architecture to the Qualcomm CTO. 2025
- SparQ Qualcomm TechFest 2026, Top 5**: presented CabinAI to the Qualcomm CTO. 2026
- DAAD Scholarship**, German Academic Exchange Service: selected for Mixed Reality research at TU Braunschweig, Germany. 2022
- CCIB SCPay 2023 Award**, Standard Chartered Bank, for the Jenkins to Azure DevOps migration. 2023
- JumpStart Hackathon**, Publicis Sapient: top performance, earned exclusive program access. 2023
- JEE Advanced**: All-India Rank **2358 / 1.2M+** (top 0.2%), Indian Institutes of Technology. 2020

LEADERSHIP

- Founding President, Kalipatnapu AR/VR Lab, BITS Pilani** (2023 – 2024): led a 40-student team and ran an Oasis VR zone (**Rs. 40,000+**).
- Blockchain Dept Coordinator, BITS Coding Club** (2022 – 2023): chartered the new department, ran recruitments, organized an **ICP Hackathon**.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, Kotlin, JavaScript, SQL, Bash, C#.

ML / LLM Stack: PyTorch, Hugging Face Transformers, LangChain, vLLM, Ollama, ONNX Runtime, RAG, LoRA / QLoRA fine-tuning, multi-agent orchestration, prompt evaluation, structured-output / tool use, embeddings (Tree-sitter, BGE, ChromaDB).

Edge / On-Device AI & Systems: Snapdragon AIC100, Hexagon NPU, Qualcomm Genie SDK + QNN, AI Hub, INT4/INT8 quantization, Llama.cpp; Linux, Docker, Kubernetes, Redis, Flask, Azure DevOps, Git, CI/CD.